

7 (a)

$$A = A_0 e^{-\lambda t}$$

$$3.7 \times 10^{11} = A_0 e^{-\frac{\ln 2}{5.2}(2.5)}$$

$$A_0 = 5.16 \times 10^{11}$$

M1

A1

(b)

$$A_0 = \lambda N_0$$
$$N_0 = \frac{5.16 \times 10^{11}}{5.2 \times 365 \times 24 \times 60 \times 60}$$
$$= 1.2216 \times 10^{20}$$

C1

C1

$$\text{mass} = \frac{1.2216 \times 10^{20}}{6.02 \times 10^{23}} \times 60 = 0.012 \text{ g}$$

A1

(c)

Rate of emission of energy

B1

$$= (3.7 \times 10^{11} \times (0.31 + 1.17 + 1.33)(1.60 \times 10^{-19}))$$

$$= 0.166 \text{ W}$$

B1